

New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
SELECT
    (SELECT COUNT(DISTINCT CUSTOMER_ID) FROM order_t) AS total_customers,
    c.STATE,
    COUNT(DISTINCT o.CUSTOMER_ID) AS customers_in_state
FROM
    order_t o
JOIN
    customer_t c ON o.CUSTOMER_ID = c.CUSTOMER_ID
GROUP BY
    c.STATE
ORDER BY
    customers_in_state DESC;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT
    (SELECT COUNT(DISTINCT CUSTOMER_ID) FROM order_t) AS total_customers,
    c.STATE,
    COUNT(DISTINCT o.CUSTOMER_ID) AS customers_in_state
FROM
    order_t o
JOIN
    customer_t c ON o.CUSTOMER_ID = c.CUSTOMER_ID
GROUP BY
    c.STATE
ORDER BY
    customers_in_state DESC
```

Output:

Showing first 10 rows out of 49 rows

total_customers	state	customers_in_state
994	Texas	97
994	California	97
994	Florida	86
994	New York	69
994	District of Columbia	35
994	Ohio	33
994	Colorado	33
994	Alabama	29
994	Washington	28
994	Arizona	26

Observations and Insights:

- A total of **994 unique customers** have placed orders.
- These customers are distributed across **49 different U.S. states**.
- The top 3 states with the highest number of customers are:
 1. **Texas:** 97 customers
 2. **California:** 97 customers
 3. **Florida:** 86 customers
- States like **New York, District of Columbia, and Ohio** also show strong engagement.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT
    p.VEHICLE_MAKER,
    COUNT(DISTINCT o.CUSTOMER_ID) AS customer_count
FROM
    order_t o
JOIN
    product_t p ON o.PRODUCT_ID = p.PRODUCT_ID
GROUP BY
    p.VEHICLE_MAKER
ORDER BY
    customer_count DESC
LIMIT 5;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT
  p.VEHICLE_MAKER,
  COUNT(DISTINCT o.CUSTOMER_ID) AS customer_count
FROM
  order_t o
JOIN
  product_t p ON o.PRODUCT_ID = p.PRODUCT_ID
GROUP BY
  p.VEHICLE_MAKER
ORDER BY
  customer_count DESC
LIMIT 5
```

Output:

Showing 5 rows

vehicle_maker	customer_count
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations and Insights:

- The top 5 vehicle makers by unique customer preference are:
 1. **Chevrolet** – 83 customers
 2. **Ford** – 63 customers
 3. **Toyota** – 52 customers
 4. **Pontiac** – 50 customers
 5. **Dodge** – 50 customers
- **Chevrolet** leads the customer preference list, with 20 more unique buyers than the next closest brand.
- **Ford** and **Toyota** continue to show strong brand recognition in the resale market.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT
    STATE,
    VEHICLE_MAKER,
    customer_count
FROM (
    SELECT
        c.STATE,
        p.VEHICLE_MAKER,
        COUNT(DISTINCT o.CUSTOMER_ID) AS customer_count,
        RANK() OVER (
            PARTITION BY c.STATE
            ORDER BY COUNT(DISTINCT o.CUSTOMER_ID) DESC
        ) AS rnk
    FROM
        order_t o
    JOIN
        customer_t c ON o.CUSTOMER_ID = c.CUSTOMER_ID
    JOIN
        product_t p ON o.PRODUCT_ID = p.PRODUCT_ID
    GROUP BY
        c.STATE, p.VEHICLE_MAKER
)
WHERE rnk = 1
ORDER BY STATE;
```



Output:

Result: Passed

Query 1

Query:

```
SELECT
  STATE,
  VEHICLE_MAKER,
  customer_count
FROM (
  SELECT
    c.STATE,
    p.VEHICLE_MAKER,
    COUNT(DISTINCT o.CUSTOMER_ID) AS customer_count,
    RANK() OVER (
      PARTITION BY c.STATE
      ORDER BY COUNT(DISTINCT o.CUSTOMER_ID) DESC
    ) AS rnk
  FROM
    order_t o
  JOIN
    customer_t c ON o.CUSTOMER_ID = c.CUSTOMER_ID
  JOIN
    product_t p ON o.PRODUCT_ID = p.PRODUCT_ID
  GROUP BY
    c.STATE, p.VEHICLE_MAKER
)
WHERE rnk = 1
ORDER BY STATE
```

Output:

Showing first 10 rows out of 143 rows

STATE	VEHICLE_MAKER	customer_count
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arizona	Cadillac	3
Arkansas	Volkswagen	1
Arkansas	Suzuki	1
Arkansas	Pontiac	1
Arkansas	Mitsubishi	1
Arkansas	GMC	1
Arkansas	Chevrolet	1

Observations and Insights:

- The result identifies the **most preferred vehicle maker** in every state, based on the **number of distinct customers**.
- In some states (e.g., **Arizona, Arkansas**), **multiple vehicle makers share the same top rank**, showing a tie in customer count.
- **Chevrolet** and **Pontiac** appear frequently across multiple states.
- In certain states (e.g., **Alaska, Arkansas**), the **top preference is based on very few customers** (sometimes only 1).

Question 4: Find the overall average rating given by the customers.

What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
SELECT
    quarter_number,
    ROUND(AVG(feedback_score), 2) AS average_rating,
    CASE ROUND(AVG(feedback_score))
        WHEN 5 THEN 'Very Good'
        WHEN 4 THEN 'Good'
        WHEN 3 THEN 'Okay'
        WHEN 2 THEN 'Bad'
        ELSE 'Very Bad'
    END AS feedback_label
FROM (
    SELECT
        quarter_number,
        CASE customer_feedback
            WHEN 'Very Bad' THEN 1
            WHEN 'Bad' THEN 2
            WHEN 'Okay' THEN 3
            WHEN 'Good' THEN 4
            WHEN 'Very Good' THEN 5
        END AS feedback_score
    FROM order_t
    WHERE customer_feedback IS NOT NULL
) AS scored_feedback
```



```
GROUP BY quarter_number  
  
ORDER BY quarter_number;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT  
  quarter_number,  
  ROUND(AVG(feedback_score), 2) AS average_rating,  
  CASE ROUND(AVG(feedback_score))  
    WHEN 5 THEN 'Very Good'  
    WHEN 4 THEN 'Good'  
    WHEN 3 THEN 'Okay'  
    WHEN 2 THEN 'Bad'  
    ELSE 'Very Bad'  
  END AS feedback_label  
FROM (  
  SELECT  
    quarter_number,  
    CASE customer_feedback  
      WHEN 'Very Bad' THEN 1  
      WHEN 'Bad' THEN 2  
      WHEN 'Okay' THEN 3  
      WHEN 'Good' THEN 4  
      WHEN 'Very Good' THEN 5  
    END AS feedback_score  
  FROM order_t  
  WHERE customer_feedback IS NOT NULL  
) AS scored_feedback  
GROUP BY quarter_number  
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	average_rating	feedback_label
1	3.55	Good
2	3.35	Okay
3	2.96	Okay
4	2.4	Bad

Observations and Insights:

- **Declining Trend:** The average customer feedback rating has been **steadily declining** across quarters.
- **Quarter 1 started strong** with a **Good** average rating of 3.55, showing healthy customer sentiment.
- **By Quarter 2 and 3**, ratings dipped into the **"Okay"** zone, indicating emerging dissatisfaction.
- **Quarter 4 is concerning**, with an average rating of **2.40 (Bad)** — a clear signal that **customer satisfaction has dropped significantly**.
- This trend may correlate with a **drop in new customer acquisition**, as mentioned in the business context.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT

    quarter_number,

    ROUND(SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1.0 ELSE 0 END) * 100.0
    / COUNT(*), 2) AS "Very Bad",

    ROUND(SUM(CASE WHEN customer_feedback = 'Bad' THEN 1.0 ELSE 0 END) * 100.0 /
    COUNT(*), 2) AS "Bad",

    ROUND(SUM(CASE WHEN customer_feedback = 'Okay' THEN 1.0 ELSE 0 END) * 100.0 /
    COUNT(*), 2) AS "Okay",

    ROUND(SUM(CASE WHEN customer_feedback = 'Good' THEN 1.0 ELSE 0 END) * 100.0 /
    COUNT(*), 2) AS "Good",

    ROUND(SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1.0 ELSE 0 END) *
    100.0 / COUNT(*), 2) AS "Very Good"

FROM order_t

WHERE customer_feedback IS NOT NULL

GROUP BY quarter_number

ORDER BY quarter_number;
```

Output:

Result: **Passed**

✓ Query 1

Query:

```
SELECT
  quarter_number,
  ROUND(SUM(CASE WHEN customer_feedback = 'Very Bad' THEN 1.0 ELSE 0 END) * 100.0 / COUNT(*), 2) AS "Very Bad",
  ROUND(SUM(CASE WHEN customer_feedback = 'Bad' THEN 1.0 ELSE 0 END) * 100.0 / COUNT(*), 2) AS "Bad",
  ROUND(SUM(CASE WHEN customer_feedback = 'Okay' THEN 1.0 ELSE 0 END) * 100.0 / COUNT(*), 2) AS "Okay",
  ROUND(SUM(CASE WHEN customer_feedback = 'Good' THEN 1.0 ELSE 0 END) * 100.0 / COUNT(*), 2) AS "Good",
  ROUND(SUM(CASE WHEN customer_feedback = 'Very Good' THEN 1.0 ELSE 0 END) * 100.0 / COUNT(*), 2) AS "Very Good"
FROM order_t
WHERE customer_feedback IS NOT NULL
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	Very Bad	Bad	Okay	Good	Very Good
1	10.97	11.29	19.03	28.71	30
2	14.89	14.12	20.23	22.14	28.63
3	17.9	22.71	21.83	20.96	16.59
4	30.65	29.15	20.1	10.05	10.05

Observations and Insights:

- In Quarter 1, Very Good (30%) and Good (28.71%) made up nearly 60% of all feedback — indicating high satisfaction.
- Positive feedback (Good + Very Good) steadily declines over each quarter, dropping to just 20.1% combined in Q4.
- "Very Bad" feedback nearly triples, from 10.97% in Q1 to 30.65% in Q4. Similarly, "Bad" feedback rises from 11.29% to 29.15%.
- More than 59% of customers gave "Bad" or "Very Bad" feedback in Q4. This suggests a serious issue with product or service quality toward the end of the year.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT  
  
    quarter_number,  
  
    COUNT(*) AS total_orders  
  
FROM order_t  
  
GROUP BY quarter_number  
  
ORDER BY quarter_number;
```

Output:

Result: Passed

✓ Query 1

Query:

```
SELECT  
    quarter_number,  
    COUNT(*) AS total_orders  
FROM order_t  
GROUP BY quarter_number  
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	total_orders
1	310
2	262
3	229
4	199

Observations and Insights:

- The number of orders **declined steadily every quarter**, from **310 in Q1** to **199 in Q4**. That's a 36% drop.
- **Q4 had the lowest order volume**, indicating either a demand drop, fulfillment issues, or possibly poor customer experience.

Question 7: Calculate the net revenue generated by the company.

What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
SELECT
    o.quarter_number,
    ROUND(SUM(o.quantity * (p.vehicle_price - o.discount)), 2) AS net_revenue,
    ROUND(
        100.0 * (
            SUM(o.quantity * (p.vehicle_price - o.discount)) -
            LAG(SUM(o.quantity * (p.vehicle_price - o.discount))) OVER (ORDER BY
o.quarter_number)
        ) / LAG(SUM(o.quantity * (p.vehicle_price - o.discount))) OVER (ORDER BY
o.quarter_number),
        2
    ) AS revenue_change
FROM order_t o
JOIN product_t p ON o.product_id = p.product_id
GROUP BY o.quarter_number
ORDER BY o.quarter_number;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT
  o.quarter_number,
  ROUND(SUM(o.quantity * (p.vehicle_price - o.discount)), 2) AS net_revenue,
  ROUND(
    100.0 * (
      SUM(o.quantity * (p.vehicle_price - o.discount)) -
      LAG(SUM(o.quantity * (p.vehicle_price - o.discount))) OVER (ORDER BY o.quarter_number)
    ) / LAG(SUM(o.quantity * (p.vehicle_price - o.discount))) OVER (ORDER BY o.quarter_number),
    2
  ) AS revenue_change
FROM order_t o
JOIN product_t p ON o.product_id = p.product_id
GROUP BY o.quarter_number
ORDER BY o.quarter_number
```

Output:

Showing 4 rows

quarter_number	net_revenue	revenue_change
1	39637378.16	
2	32913497.49	-16.96
3	29435188.49	-10.57
4	23495814.02	-20.18

Observations and Insights:

- Strong Start in Q1 - Net revenue peaked at nearly 40 million in Q1 — indicating strong sales momentum early in the year.
- Consistent Decline Each Quarter** - Each subsequent quarter shows a **steady drop** in net revenue:
 - Q2:** -16.96%
 - Q3:** -10.57%
 - Q4:** -20.18%
- The sharpest decline came in **Q4**, where revenue dropped at -20.18% compared to Q3.
- This trend corresponds with falling **order counts** and **feedback ratings** in previous questions suggesting that customer experience might be driving the drop.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT
    o.quarter_number,
    ROUND(SUM(o.quantity * (p.vehicle_price - o.discount)), 2) AS net_revenue,
    COUNT(*) AS total_orders
FROM order_t o
JOIN product_t p ON o.product_id = p.product_id
GROUP BY o.quarter_number
ORDER BY o.quarter_number;
```

Output:

Result: Passed

✓ Query 1

Query:

```
SELECT
    o.quarter_number,
    ROUND(SUM(o.quantity * (p.vehicle_price - o.discount)), 2) AS net_revenue,
    COUNT(*) AS total_orders
FROM order_t o
JOIN product_t p ON o.product_id = p.product_id
GROUP BY o.quarter_number
ORDER BY o.quarter_number
```

Output:

Showing 4 rows

quarter_number	net_revenue	total_orders
1	39637378.16	310
2	32913497.49	262
3	29435188.49	229
4	23495814.02	199

Observations and Insights:

- Both **net revenue** and **order volume** declined consistently each quarter. Suggests a strong **correlation between order count and revenue** — fewer orders directly impacted revenue.
- Q1 recorded the **highest** number of orders and revenue, indicating a strong start for the year.

- In Q4, both metrics hit their **lowest**:
 - Revenue dropped by ~41% from Q1
 - Orders dropped by ~36%

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
    c.credit_card_type,
    ROUND(AVG(o.discount), 2) AS average_discount
FROM order_t o
JOIN customer_t c ON o.customer_id = c.customer_id
GROUP BY c.credit_card_type
ORDER BY average_discount DESC;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT
    c.credit_card_type,
    ROUND(AVG(o.discount), 2) AS average_discount
FROM order_t o
JOIN customer_t c ON o.customer_id = c.customer_id
GROUP BY c.credit_card_type
ORDER BY average_discount DESC
```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	average_discount
laser	0.64
mastercard	0.63
visa-electron	0.62
maestro	0.62
instapayment	0.62
china-unionpay	0.62
americanexpress	0.62
visa	0.61
visa-prepaid	0.61
visa-gift	0.61

Observations and Insights:

- There is minimal variation. All credit card types received an **average discount between 0.61 and 0.64**.
- Customers using **Laser cards** got the **highest average discount (0.64)**, followed by **Mastercard (0.63)**.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT

    quarter_number,

    ROUND(AVG(julianday(ship_date) - julianday(order_date)), 2) AS
    avg_shipping_days

FROM order_t

WHERE ship_date IS NOT NULL AND order_date IS NOT NULL

GROUP BY quarter_number

ORDER BY quarter_number;
```

Output:

Result: Passed

Query 1

Query:

```
SELECT
    quarter_number,
    ROUND(AVG(julianday(ship_date) - julianday(order_date)), 2) AS avg_shipping_days
FROM order_t
WHERE ship_date IS NOT NULL AND order_date IS NOT NULL
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	avg_shipping_days
1	57.17
2	71.11
3	117.76
4	174.1

Observations and Insights:

- There is steep increase in Shipping Time. Average shipping duration tripled from Q1 (57 days) to Q4 (174 days). This suggests significant delays in order fulfillment and logistics over the year.
- Crossing **100+ days** in Q3 and **174+ days** in Q4 is alarming — indicates potential **supply chain issues, inventory delays, or operational inefficiencies**.

- Strong Correlation with Customer Dissatisfaction. This delay aligns with increasing "Very Bad" and "Bad" feedback and the drop in orders and revenue.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
1,25,78,078.16	1,000	758	3.36
Last Quarter Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
23,495,814.02	199	174.10	10.05%

Business Recommendations

KEY INSIGHTS:

- **Declining Orders and Revenue**
 - Order volume dropped from **310 in Q1** to **199 in Q4** (↓36%).
 - Revenue also declined from **₹3.96 Cr** to **₹2.34 Cr** (↓40%).
 - Indicates possible loss of customer interest, repeat business, or demand.
- **Rising Shipping Delays**
 - Average shipping time **tripled** from **57 days in Q1** to **174 days in Q4**.
 - Suggests serious **operational inefficiencies**, possibly in logistics or inventory management.
- **Decreasing Customer Satisfaction**
 - Average rating is 3.36, and Q4 showed the lowest satisfaction.
 - % of "Good" feedback dropped significantly over time.
 - Indicates a **decline in service quality**, likely driven by long wait times and poor fulfillment.
- **Customer Distribution**
 - Business serves customers across multiple states, but **some states have much higher engagement**.
 - **Top vehicle makers** and **regional preferences** offer a great opportunity for localized marketing.
- **Uniform Discount Strategy**
 - Discounts offered across all credit card types are similar (~0.61 to 0.64).
 - May be effective for consistency but misses the chance for targeted promotions.

RECOMMENDATIONS:

- **Fix Operational Bottlenecks Immediately**
 - Need to investigate why shipping delays have grown so significantly.
 - Need to optimize supply chain, vendors, inventory, and logistics.
- **Improve Customer Experience**
 - Shorten delivery times to rebuild customer trust.
 - Possibly implement **real-time tracking, proactive updates**, and customer support.
 - Actively monitor and respond to feedback to recover brand perception.
- **Boost Retention & Re-engagement**
 - Q1 performance shows strong potential — use loyalty programs and re-targeting for lapsed customers.
 - Focus on **top-performing states and vehicle makers** for tailored campaigns.
- **Experiment with Differentiated Discounts**
 - Introduce personalized offers based on card type, geography, or purchase behavior.
 - Collaborate with banks to launch **co-branded card offers** or seasonal incentives.
- **Track KPIs Quarterly**
 - Monitor revenue, shipping time, and feedback score side-by-side.
 - Set targets to bring **shipping <60 days** and **ratings >4.0** in the next quarter.
- The business shows potential but is clearly facing **customer dissatisfaction and operational issues** that have **cascaded into falling revenue and order volume**.
- By realigning logistics, personalizing marketing, and focusing on customer satisfaction, the company can **stabilize and return to Q1-level performance** in the coming year.